

Large-Model Prior and Causal Consistency for Unseen-Condition Multimodal Defect Recognition in Additive Manufacturing

Junwei Gu, Jiahui Guo, and Yu Wang

Abstract—In-situ monitoring in metal additive manufacturing typically provides multimodal observations such as visual and infrared signals. When training and test conditions differ, however, defect recognition models can mistake condition-related cues, including background texture and thermal-field distribution, for defect evidence, resulting in substantial performance degradation under unseen conditions. To address this problem, this paper proposes the Multimodal Causal-Stable Qwen Network (MCSQ-Net). The method uses the visual encoding component of a pretrained large model as a shared feature extractor, constructs a unified fusion space through multimodal representation learning, and imposes a causal consistency constraint on same-class samples from different conditions in the fused representation space to suppress condition-induced drift. A lightweight classification head is then used for defect category recognition, enabling task adaptation under limited supervision while preserving the generalization prior of the pretrained model. Experiments are conducted under both source-domain condition-held-out cross-validation and external transfer settings. MCSQ-Net achieves a Macro-F1 of 0.8547 and a Balanced Accuracy of 0.8556 on Task-1 defect category recognition, and a Macro-F1 of 0.6067 on Task-3 external transfer, outperforming both Prompt-Qwen and classical multimodal baselines. Ablation studies further show that the causal consistency constraint consistently improves cross-condition representation quality, whereas aggressive unfreezing of the visual backbone weakens unseen-condition generalization. These results indicate that, for industrial small-sample multimodal defect recognition, preserving large-model visual priors and explicitly learning cross-condition stable representations in the fusion space is an effective strategy for improving recognition performance under unseen conditions.

Index Terms—Selective laser melting, defect recognition, multimodal fusion, unseen-condition generalization, external transfer, large-model prior, causal consistency constraint.

I. INTRODUCTION

METAL additive manufacturing constructs complex parts layer by layer through the melting of metal powders and has been widely adopted in aerospace, medical devices, and tooling. Selective laser melting (SLM), as a representative process, involves complex physical phenomena such as melt-pool dynamics, heat transfer, and solidification [1]. Variations in process parameters can directly lead to defects including pores, cracks, and geometric distortion [2]. Even minor process deviations may substantially affect the mechanical performance and service reliability of the final part.

Industrial systems commonly deploy visible-light cameras and infrared imagers to monitor metal additive manufactur-

ing processes in a synchronized manner [3], [4]. Visible imaging captures melt-pool morphology and surface texture, whereas infrared imaging reflects heat-input distribution and thermal-field evolution. Such multimodal observations provide complementary defect evidence. Existing monitoring methods, however, generally assume that training and test data follow the same distribution. When process parameters, equipment models, or powder batches change, recognition performance often degrades significantly.

This failure mode is rooted in the tendency of deep models to rely on spurious correlations. Instead of learning causal defect characteristics, the model may exploit superficial statistical cues such as background texture, illumination patterns, or device noise. In multimodal monitoring for metal additive manufacturing, this means that the model may incorrectly treat condition-specific thermal backgrounds as defect evidence, which leads to failure under unseen conditions. Learning stable recognition rules that transfer from seen training conditions to unseen test conditions is therefore a central problem in defect recognition for metal additive manufacturing, particularly SLM.

Pretrained large visual models offer a possible solution. Trained on massive visual and multimodal corpora, these models provide strong priors for handling heterogeneous inputs and, in principle, should help extract condition-stable representations. Existing evidence nevertheless shows that simple prompt-based adaptation is inadequate under unseen-condition settings. Large-model priors do not automatically translate into accurate industrial defect recognition. The key challenge is how to adapt them in a way that preserves cross-condition stability while effectively integrating complementary information from multiple sensors.

This paper proposes the Multimodal Causal-Stable Qwen Network (MCSQ-Net) for unseen-condition defect recognition in metal additive manufacturing. The central idea is to use a pretrained large visual model as a cross-condition feature extractor, explicitly fuse visible and infrared information, and impose a causal consistency constraint in the fusion space. MCSQ-Net adopts a lightweight discriminative adaptation strategy so that cross-condition invariant defect representations can be learned while retaining the pretrained prior. Specifically, the causal stability assumption is that condition-related cues such as texture, brightness, and thermal background vary with process settings, whereas essential defect characteristics should remain stable across conditions. By constraining same-class samples from different conditions to be close in the

fusion space, the model is driven to reduce its reliance on unstable environmental factors and to learn a more robust mapping from seen to unseen conditions.

The main contributions are threefold.

- 1) We verify the necessity of multimodal fusion for unseen-condition defect recognition. By comparing single-modality and multimodal configurations under cross-condition evaluation, we show that combining visible and infrared information improves adaptation to unseen devices and unseen process settings, thereby providing empirical guidance for sensor configuration in industrial monitoring systems.
- 2) We develop a large-model-prior-guided framework for unseen-condition defect recognition. The framework freezes the pretrained visual backbone and introduces a learnable multimodal fusion head for lightweight discriminative adaptation, balancing deployment efficiency and recognition accuracy. Experiments show that this strategy maintains stable detection performance in cross-device transfer scenarios.
- 3) We introduce a causal-stable representation learning mechanism. By enforcing consistency across same-class samples from different conditions, the model is encouraged to suppress background texture, thermal noise, and other environment-dependent cues, and to focus instead on the intrinsic characteristics of the defect itself. This mechanism improves generalization from seen to unseen conditions and provides a practical route toward robust industrial vision monitoring.

II. RELATED WORK

A. Multimodal Defect Monitoring in Metal Additive Manufacturing

In-situ monitoring for metal additive manufacturing has evolved along three main directions: visible-light imaging, infrared thermography, and multi-sensor fusion [1]–[4]. Existing studies have consistently shown that multimodal observations can characterize defect formation from complementary perspectives, including morphology, thermal input, and process dynamics, and therefore are more informative than single-modality sensing.

Representative studies have validated the effectiveness of multimodal monitoring. Wu *et al.* [11] demonstrated that multi-sensor monitoring improves quality-level recognition. Liu *et al.* [12] reconstructed melt-pool visual characteristics from acoustic and thermal-radiation signals, offering a lower-cost alternative to high-speed imaging. Shen *et al.* [13] jointly modeled infrared images, top-view images, process parameters, and physical simulation results, while Li *et al.* [14] discussed the transferability of multimodal features across materials. These studies confirm the value of multimodal fusion in defect monitoring, but most still assume that training and test data are drawn from similar distributions. Less attention has been paid to the distribution shift jointly induced by changes in equipment, process parameters, and operating environments under unseen conditions. In contrast, this paper focuses on learning multimodal defect representations that remain stable under condition-disjoint evaluation.

B. Large-Model Adaptation and Cross-Domain Generalization

Pretrained large models provide a new source of prior knowledge for industrial anomaly and defect recognition. Recent work such as AnomalyGPT [5], Myriad [6], OmniAD [9], and AD-Copilot [10] has demonstrated the potential of large models in industrial vision through prompt learning, expert-model integration, supervised fine-tuning, and visual context comparison. At the same time, lightweight adaptation strategies such as QLoRA and active learning [7], [8] indicate that task specialization can be achieved in resource-constrained settings without full-parameter fine-tuning.

However, existing methods mainly address anomaly detection, localization, or few-shot adaptation within a single dataset, and rarely investigate the cross-domain generalization problem caused jointly by modality heterogeneity and condition shift in metal additive manufacturing. When training and test conditions are strictly disjoint, prompt-based or conventional fine-tuning strategies are often insufficient to ensure that the model learns genuinely stable defect evidence. Different from these studies, this paper uses the visual prior of a pretrained large model for multimodal unified encoding and introduces a cross-condition causal consistency constraint in the fusion space to explicitly improve defect recognition stability under unseen conditions.

III. UNSEEN-CONDITION TASK FORMULATION

Unseen-condition multimodal defect recognition in metal additive manufacturing can be formulated as a supervised classification problem under conditional extrapolation. For sample i , let the joint observation be $\mathbf{x}_i = \{\mathbf{x}_i^{(k)} \mid k \in \mathcal{M}_i\} \in \mathcal{X}$, where $\mathcal{M}_i$ denotes the set of available modalities. For visual and infrared observations, $\mathbf{x}_i^{(k)} \in \mathbb{R}^{H \times W \times C}$ denotes the input from the k th sensor, $y_i \in \mathcal{Y}$ is the defect label, $c_i \in \mathcal{C}$ is the condition identifier, and $\mathcal{X}$ is the multimodal observation space. The condition variable c_i is used to represent external factors that change the observation distribution, including equipment configuration, process parameters, printing task, and material state. Unlike conventional in-distribution classification, the training set $\mathcal{D}_{\text{train}} = \{(\mathbf{x}_i, y_i, c_i)\}_{i=1}^N$ and the test set $\mathcal{D}_{\text{test}}$ come from condition sets $\mathcal{C}_{\text{train}}$ and $\mathcal{C}_{\text{test}}$, respectively, with $\mathcal{C}_{\text{train}} \cap \mathcal{C}_{\text{test}} = \emptyset$. The model therefore faces not only new instances at test time, but also systematic distribution shifts caused by changes in equipment, process settings, and task conditions.

Under this setting, the model must learn a classifier $f : \mathcal{X} \rightarrow \mathcal{Y}$ solely from $\mathcal{D}_{\text{train}}$ while maintaining low risk on unseen conditions:

$$\mathcal{R}_{\text{uc}}(f) = \mathbb{E}_{(\mathbf{x}, y, c) \sim P_{\text{test}}} [\ell(f(\mathbf{x}), y)], \quad c \in \mathcal{C}_{\text{test}}, \quad (1)$$

where $\ell(\cdot, \cdot)$ is the classification loss, $\mathcal{R}_{\text{uc}}$ denotes the unseen-condition risk, and P_{test} is the test-time data distribution. This objective is difficult because multimodal observations simultaneously contain stable discriminative factors related to defect formation and unstable background statistics or sensor-response differences induced by condition changes. Direct empirical risk minimization can exploit the latter to fit the

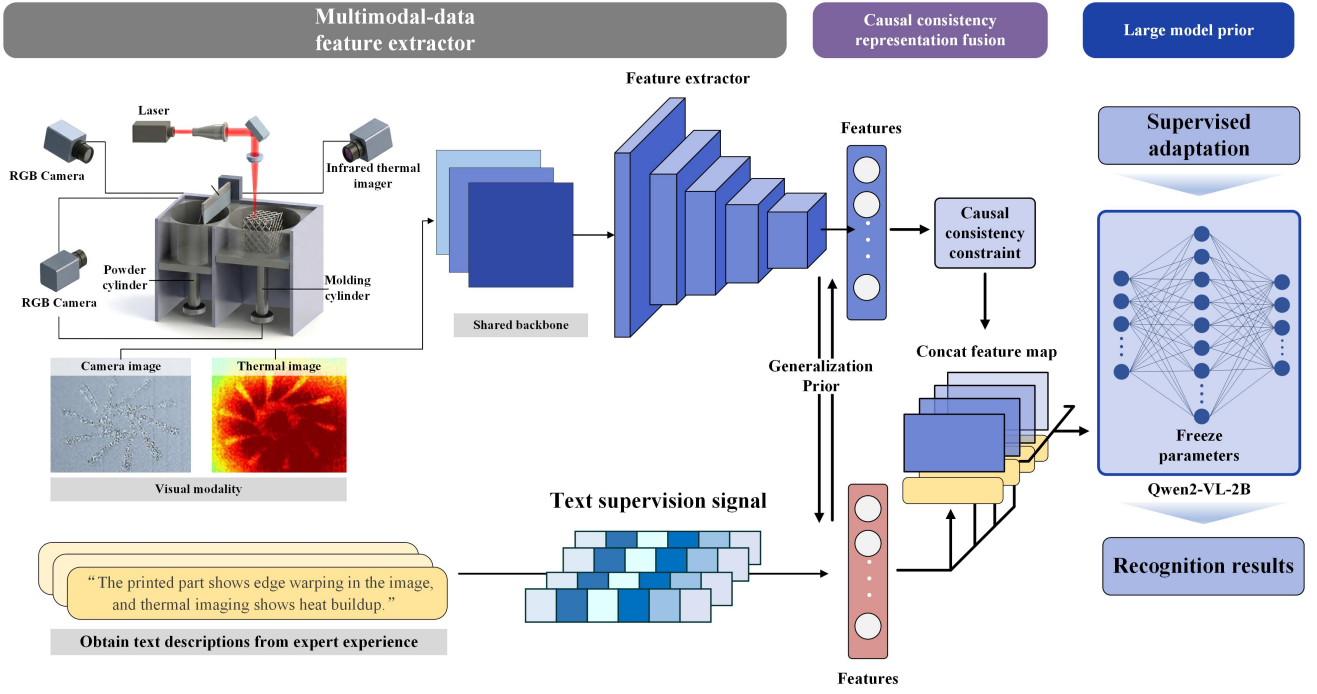


Fig. 1. Overview of MCSQ-Net. The pipeline consists of shared visual encoding, multimodal causal-consistent fusion, and lightweight adaptation guided by large-model generalization priors.

training set, thereby encoding condition-dependent superficial correlations into the decision rule and causing failure on C_{test} . Based on this task definition, the proposed method addresses three aspects: unified representation of heterogeneous observations, extraction of condition-stable features, and transfer of large-model priors.

IV. PROPOSED METHOD

To address the difficulties of heterogeneous multimodal representation, condition-dependent decision drift, and limited extrapolation under scarce labeled data, this paper proposes MCSQ-Net. For sample i , let its effective modality set be $\mathcal{M}_i = \{m_1, \dots, m_{K_i}\}$, where K_i is the number of modalities involved in fusion. Rather than using a pretrained large model directly for prompt-based closed-set classification, MCSQ-Net takes its visual encoding component as a shared feature extractor and successively performs multimodal representation learning, cross-condition consistent fusion, and lightweight discriminative adaptation. In this way, the method seeks to suppress condition-induced representation drift while preserving as much of the pretrained prior as possible. The overall framework is shown in Fig. 1.

A. Multimodal Feature Extraction

Different sensors exhibit different imaging mechanisms and statistical properties. Training a separate encoder for each modality can easily cause the model to memorize condition-related textures under limited supervision rather than stable defect cues. To mitigate this issue, MCSQ-Net shares a single pretrained large-model visual backbone f_{θ_v} across all modalities, first mapping heterogeneous observations into a

unified semantic space and then constructing modality-level representations within that space.

For any modality $m \in \mathcal{M}_i$, let the input observation be $\mathbf{x}_i^{(m)}$. The shared visual encoder produces the spatial feature sequence

$$\mathbf{E}_i^{(m)} = f_{\theta_v}(\mathbf{x}_i^{(m)}) = \left[\mathbf{e}_{i,1}^{(m)}, \dots, \mathbf{e}_{i,P_i^{(m)}}^{(m)} \right]^\top \in \mathbb{R}^{P_i^{(m)} \times d_v}, \quad (2)$$

where d_v is the feature dimension, $P_i^{(m)}$ is the number of retained local features after spatial compression in the visual backbone, and $\mathbf{e}_{i,p}^{(m)}$ is the local visual feature at the p th spatial position. Since the downstream fusion module requires fixed-length inputs, the spatial feature sequence is mean-pooled to obtain a modality-level representation:

$$\mathbf{z}_i^{(m)} = \frac{1}{P_i^{(m)}} \sum_{p=1}^{P_i^{(m)}} \mathbf{e}_{i,p}^{(m)} \in \mathbb{R}^{d_v}. \quad (3)$$

Here, $\mathbf{z}_i^{(m)}$ is the global semantic vector of modality m . This operation compresses a variable-length spatial sequence into a fixed-length vector and thereby reduces scale differences across modalities. For all input modalities of sample i , the modality-level feature group is

$$\mathbf{Z}_i = \left[\mathbf{z}_i^{(m_1)}, \dots, \mathbf{z}_i^{(m_{K_i})} \right] \in \mathbb{R}^{K_i \times d_v}. \quad (4)$$

where K_i is the number of modalities actually involved in fusion. In the main setting, the shared visual backbone remains frozen and is used only as a unified feature extractor. This choice preserves the pretrained visual prior and provides a more stable basis for the subsequent cross-condition consistency constraint.

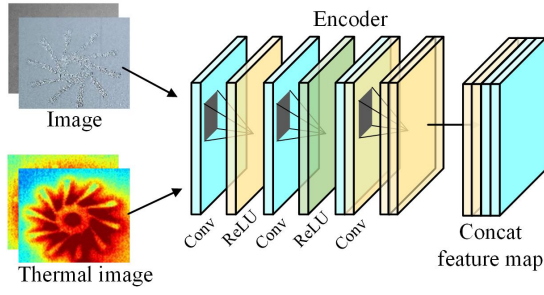


Fig. 2. Multimodal feature extraction module.

B. Causal-Consistent Representation Fusion

After obtaining modality-level vectors, directly sending them into a classifier would still allow the model to encode brightness bias, thermal background, and texture noise as decision cues. The goal of this stage is therefore not to design an increasingly complex fusion network, but to explicitly constrain condition-stable factors at the fused representation level so that the representation is more tightly tied to the defect itself.

Specifically, the modality vectors of sample i are first concatenated along the feature dimension to form the joint description

$$\mathbf{g}_i = \text{Concat}(\mathbf{z}_i^{(m_1)}, \dots, \mathbf{z}_i^{(m_{K_i})}) \in \mathbb{R}^{K_i d_v}. \quad (5)$$

The fused representation is then obtained by mapping this joint description into a common fusion space:

$$\mathbf{h}_i = \phi_{\theta_f}(\text{LN}(\mathbf{g}_i)) \in \mathbb{R}^{d_h}. \quad (6)$$

where $\text{LN}(\cdot)$ denotes LayerNorm and d_h is the fusion-space dimension. The fusion head ϕ_{θ_f} is only responsible for feature compression and cross-modal recombination; suppression of condition-dependent spurious correlations is mainly achieved by the subsequent consistency constraint.

The causal consistency constraint is built on the assumption that, for the same defect category, background texture, brightness distribution, and thermal response may vary across conditions, whereas defect morphology and abnormal thermal signatures that determine class identity should remain stable. To make this stable component comparable in the fusion space, the fused vector is first ℓ_2 -normalized:

$$\bar{\mathbf{h}}_i = \frac{\mathbf{h}_i}{\|\mathbf{h}_i\|_2} \in \mathcal{S}^{d_h-1}. \quad (7)$$

The normalized vector $\bar{\mathbf{h}}_i$ retains only directional information. Consequently, its inner product with another normalized vector directly corresponds to cosine similarity, which is more suitable for measuring semantic consistency across conditions.

For each sample, a same-class but different-condition reference set is constructed. For sample i , define the cross-condition positive pool as

$$\Omega_i = \{j \mid j \neq i, y_j = y_i, c_j \neq c_i\}. \quad (8)$$

Here, $y_j = y_i$ ensures that the two samples belong to the same defect category, and $c_j \neq c_i$ ensures that they come from different conditions. The resulting consistency relation is

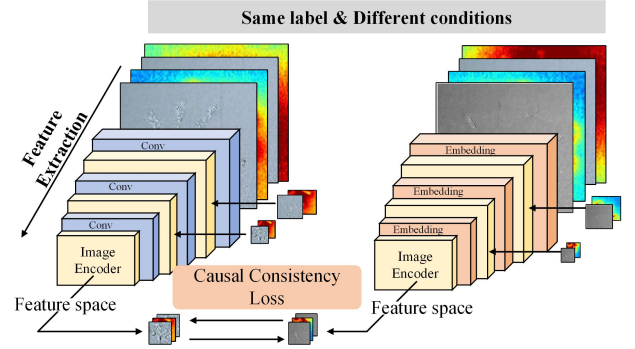


Fig. 3. Causal-consistent fusion module.

therefore defined between sample-level cross-condition positive pairs rather than between different modalities. If a sample does not have an available positive pair, it does not contribute to the consistency loss in the current mini-batch.

For the subset of anchors in a mini-batch $\mathcal{B}$ that have valid positives, denoted by $\mathcal{B}^+ = \{i \in \mathcal{B} \mid \Omega_i \neq \emptyset\}$, one positive $\pi(i)$ is sampled from Ω_i . The causal consistency loss is then written as

$$\mathcal{L}_{cc} = \frac{1}{|\mathcal{B}^+|} \sum_{i \in \mathcal{B}^+} (1 - \bar{\mathbf{h}}_i^\top \text{sg}(\bar{\mathbf{h}}_{\pi(i)})), \quad (9)$$

where $\text{sg}(\cdot)$ denotes the stop-gradient operator used to block gradient backpropagation through the reference branch. Since both $\bar{\mathbf{h}}_i$ and $\bar{\mathbf{h}}_{\pi(i)}$ are normalized, their inner product directly corresponds to cosine similarity. Thus, the role of (9) is to pull together fused representations of same-class samples from different conditions, keeping consistent defect semantics close while reducing the interference of condition differences on the decision boundary.

C. Lightweight Adaptation Guided by Large-Model Priors

After obtaining the fused representation, the generalization prior formed by the pretrained large model on open-domain visual data still needs to be converted into category-specific defect discrimination. Direct end-to-end fine-tuning of the visual backbone would easily rewrite that prior into local experience tied to a small number of training conditions, whereas no task adaptation at all would make it difficult to form stable category boundaries. Therefore, under the proposed setting, the visual backbone remains frozen and only the fusion head and classification head are updated, yielding lightweight task adaptation with a small number of trainable parameters.

Concretely, the classification score vector is computed from the fused representation $\mathbf{h}_i$ as

$$\ell_i = \mathbf{W}_c \mathbf{h}_i + \mathbf{b}_c \in \mathbb{R}^{|\mathcal{Y}|}, \quad (10)$$

where $\mathbf{W}_c \in \mathbb{R}^{|\mathcal{Y}| \times d_h}$ and $\mathbf{b}_c \in \mathbb{R}^{|\mathcal{Y}|}$ are the weight matrix and bias vector of the classifier, respectively, $|\mathcal{Y}|$ is the number of defect categories, and $\ell_{i,k}$ is the unnormalized score of sample i for class k . The classification loss is standard cross-entropy:

$$\mathcal{L}_{cls} = -\frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} \log \frac{\exp(\ell_{i,y_i})}{\sum_{k=1}^{|\mathcal{Y}|} \exp(\ell_{i,k})}, \quad (11)$$

where $|\mathcal{B}|$ denotes the batch size. $\mathcal{L}_{\text{cls}}$ learns class-separating decision boundaries, whereas $\mathcal{L}_{\text{cc}}$ suppresses condition-induced representation drift; the two objectives correspond to class separability and cross-condition stability, respectively.

In the parameter update strategy, the pretrained visual backbone is frozen, and only the fusion-head parameters θ_f and the classifier parameters $\theta_c = (\mathbf{W}_c, \mathbf{b}_c)$ are updated. This lightweight adaptation strategy confines task-specific updates to a small set of added modules and thus preserves the large-model prior as much as possible under limited supervision.

Combining the above components, the final optimization objective is

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda \mathcal{L}_{\text{cc}}, \quad (12)$$

where λ is the weight of the consistency constraint. Equation (12) indicates that optimization does not only pursue class separability on the training set, but also requires the fused representation to remain stable under condition variation.

D. Training Procedure and Overall Architecture

Training follows standard mini-batch gradient descent. Each update consists of four steps: shared visual encoding, multimodal fusion, cross-condition consistency alignment, and discriminative optimization. The shared visual backbone first extracts modality-level vectors, the fusion head produces $\mathbf{h}_i$ and $\tilde{\mathbf{h}}_i$, and then $\mathcal{L}_{\text{cls}}$ and $\mathcal{L}_{\text{cc}}$ are jointly computed to update the trainable parameters. Table I presents the training pseudocode, and Table II summarizes the key architectural configuration. Detailed training hyperparameters are deferred to the experimental settings.

V. EXPERIMENTS AND ANALYSIS

A. Experimental Settings

The experimental data were collected from a self-built open SLM in-situ monitoring platform, as illustrated in Fig. 4. The platform consists of a 500 W single-mode continuous-wave fiber laser, a galvanometer scanning system, and PBAM programmable control software, allowing real-time adjustment of key process parameters such as laser power and scanning speed. The build process is conducted under high-purity argon protection, with oxygen concentration controlled below 0.1%. To acquire multimodal process signals, the platform is equipped with two KM2010-3660 industrial cameras and one Optris PI 08M short-wave infrared imager, which record top-view and side-view visible images as well as vertical infrared thermal fields, respectively, with frame-level synchronization achieved through external TTL triggering. The resulting dataset contains visual modalities, the infrared modality, and corresponding condition information, and is used for unseen-condition defect recognition evaluation.

The proposed method uses the visual backbone of Qwen2-VL-2B-Instruct, with the fusion-space dimension set to 512, dropout set to 0.1, batch size set to 4, the AdamW optimizer, a learning rate of 3×10^{-4} , weight decay of 10^{-4} , cosine scheduling, mixed-precision training, and early stopping with a patience of 2. The main experiments compare visual, infrared, and multimodal input configurations, and the best-performing

TABLE I
MCSQ-NET TRAINING PROCEDURE

Training Procedure

- 1: **Input:** training set $\mathcal{D}_{\text{train}}$, validation set $\mathcal{D}_{\text{val}}$, shared visual backbone f_{θ_v} , fusion head ϕ_{θ_f} , classifier $(\mathbf{W}_c, \mathbf{b}_c)$, and weight λ
- 2: **Initialization:** freeze all visual layers of f_{θ_v} ; build same-class cross-condition positive pools for training samples according to labels and condition_uid
- 3: **for** epoch = 1 to E **do**
- 4: **for** mini-batch $\mathcal{B}$ **do**
- 5: Read the input observations of each sample according to the current modality configuration and pass them through the shared visual backbone
- 6: Mean-pool the spatial feature sequence of each modality to obtain modality-level vectors $\mathbf{z}_i^{(m)}$
- 7: Concatenate all modality vectors and pass them through the fusion head to obtain $\mathbf{h}_i$ and its normalized form $\tilde{\mathbf{h}}_i$
- 8: Compute classification scores with the classifier and obtain the classification loss $\mathcal{L}_{\text{cls}}$
- 9: **if** valid positive pairs exist in the batch **then**
- 10: Compute the consistency loss $\mathcal{L}_{\text{cc}}$ according to the same-class cross-condition matching relation
- 11: **else**
- 12: Set $\mathcal{L}_{\text{cc}} \leftarrow 0$
- 13: **end if**
- 14: Compute the total loss $\mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda \mathcal{L}_{\text{cc}}$
- 15: Backpropagate and update the trainable parameters
- 16: **end for**
- 17: Evaluate the current model on the validation set and save the best checkpoint according to the main metric
- 18: **if** no improvement over several epochs **then**
- 19: Trigger early stopping
- 20: **end if**
- 21: **end for**
- 22: **Output:** the best model parameters on the validation set and the corresponding test results

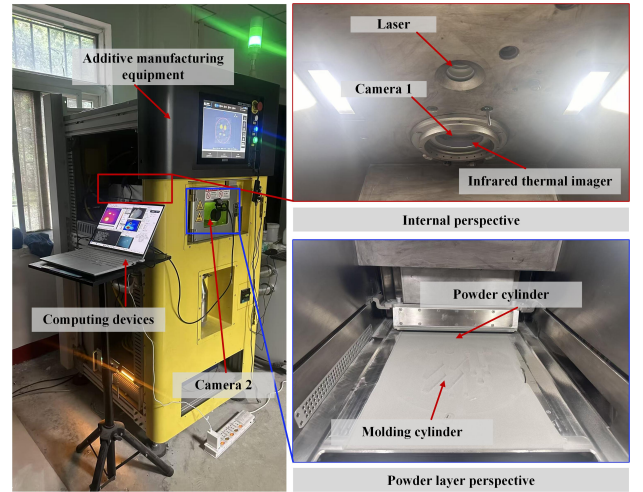


Fig. 4. Open SLM in-situ monitoring platform and multi-sensor acquisition system. The final version will show the laser processing unit, the control system, and the dual-vision and infrared sensing array.

configuration is adopted for the primary setting. Recorded process parameters are used to partition conditions and to construct same-class cross-condition positives. Category labels serve only as supervision during training and are not used as inputs during inference. The causal consistency weight is set to 0.1. The full fine-tuning comparison uses batch size 1, 3 epochs, patience 1, a visual-side learning rate of 2×10^{-6} , a

TABLE II
KEY ARCHITECTURAL CONFIGURATION OF MCSQ-NET

Module	Structure	Role
Shared visual backbone	Pretrained visual encoder, $d_v = 1536$, all visual layers frozen	Extract unified modality features
Modality aggregation	Mean pooling, $\mathbf{z}_i^{(m)} \in \mathbb{R}^{d_v}$	Generate modality-level representations
Feature concatenation	$\mathbf{g}_i = \text{Concat}(\mathbf{z}_i^{(m_1)}, \dots, \mathbf{z}_i^{(m_{K_i})}) \in \mathbb{R}^{K_i d_v}$	Preserve complementary multimodal information
Fusion head	LN+Linear($K_i d_v, 512$)+GELU+Dropout(0.1)+Linear(512, 512)+GELU	Compress and reorganize fused features
Consistency constraint	Same-label cross-condition pairing, cosine consistency loss, $\lambda = 0.1$	Suppress condition-induced shift
Classifier	Linear(512, $ \mathcal{Y} $)	Output defect-class scores
Lightweight adaptation	Freeze the visual backbone and train only $\{\theta_f, \theta_c\}$	Preserve the large-model generalization prior

TABLE III
EXPERIMENTAL TASK SETTINGS

ID	Task Name	Task Setting
Task-1	Source-domain unseen-condition defect category recognition	Three-fold cross-validation with condition-disjoint train/validation/test splits; labels are healthy (normal), high-energy warping (HEW), and low-energy abnormal lack-of-fusion (LEL); used to evaluate fine-grained defect category recognition under unseen conditions.
Task-2	Source-domain unseen-condition anomaly recognition	Same condition split as Task-1; binary classification between healthy (normal) and defective (abnormal) samples; used as a complementary task with open-set relevance to evaluate the separation of abnormal patterns.
Task-3	External-transfer defect category recognition	Trained on the source-domain training set of Task-1 and evaluated on an independent external test set; the external test set is not used for model selection; simultaneous shifts exist in device conditions, process-parameter ranges, and defect distribution; used to evaluate recognition stability under compound distribution shift.

fusion-head learning rate of 5×10^{-5} , and gradient clipping at 0.5. The strongest classical baseline is ResNet18 with dual-camera input and MLP-concat fusion. All source-domain results are aggregated over three folds, and the task settings are summarized in Table III.

We report Accuracy, Balanced Accuracy, and Macro-F1 throughout the paper. Balanced Accuracy and Macro-F1 are the primary metrics because they better reflect class-recall balance and cross-class recognition stability, both of which are critical under unseen-condition evaluation.

For clarity of presentation, the main baselines are abbreviated as follows:

- C1: ResNet18 + Camera 1, representing the single-visual-modality baseline.
- IR-Only: ResNet18 + infrared only, used to evaluate the standalone discriminative value of thermal information.
- Dual: ResNet18 + Camera 1 & 2 + MLP-concat, representing explicit feature concatenation under dual-camera input and also the strongest classical CNN baseline.
- Tri-MLP: ResNet18 + Camera 1 & 2 + IR + MLP-concat, used to test direct feature concatenation under three modalities.
- Tri-CA: ResNet18 + Camera 1 & 2 + IR + Cross-Attention, used to test explicit cross-modal interaction under three modalities.
- Tri-LF: ResNet18 + Camera 1 & 2 + IR + Late Fusion, used to evaluate decision-level fusion under three modalities.
- Prompt-Qwen: a multimodal prompt-based classification baseline that directly performs closed-set recognition from the joint input of Camera 1, Camera 2, and IR.
- MCSQ-Net: the proposed method, which uses the same

joint input of Camera 1, Camera 2, and IR but further introduces a cross-condition causal stability constraint in the fusion space.

B. Defect Recognition Under Unseen Conditions

Table IV reports the source-domain unseen-condition defect category recognition results. Among the classical CNN baselines, the single-visual model C1 already reaches a Macro-F1 of 0.8269, and the dual-camera baseline Dual further improves it to 0.8323, indicating that complementary viewpoints can improve defect discrimination. By contrast, Tri-MLP, Tri-CA, and Tri-LF do not consistently outperform Dual, suggesting that simply increasing the number of modalities or the complexity of the fusion mechanism does not automatically produce cross-condition gains. Under the same three-modality input, the two large-model-based methods diverge sharply: Prompt-Qwen reaches only 0.3914 Macro-F1, while MCSQ-Net improves the score to 0.8547 and simultaneously achieves the highest Accuracy and Balanced Accuracy. In conjunction with the trainable parameter counts listed in the table, this result indicates that the gain does not come from a larger task-specific adaptation scale, but from preserving the large-model visual prior while suppressing condition-induced drift through same-class cross-condition alignment.

Task-2 is summarized in Fig. 5. On both Balanced Accuracy and Macro-F1, the best classical baseline Dual clearly outperforms Prompt-Qwen and MCSQ-Net, indicating that anomaly recognition with open-set relevance still relies more heavily on explicit discriminative representation learning under the current setting. Even so, MCSQ-Net remains better than Prompt-Qwen on both metrics, suggesting that the causal consistency

TABLE IV
TASK-1 RESULTS ON UNSEEN-CONDITION DEFECT CATEGORY RECOGNITION

Setting	Accuracy	Bal. Acc.	Macro-F1	Trainable Parameters	Relative Inference Cost
C1	0.8237 ± 0.0440	0.8348 ± 0.0429	0.8269 ± 0.0483	11.18M	Low
IR-Only	0.7271 ± 0.1127	0.7298 ± 0.1160	0.7117 ± 0.1254	11.18M	Low
Tri-MLP	0.8138 ± 0.1127	0.8135 ± 0.1131	0.8106 ± 0.1143	12.23M	Medium
Tri-CA	0.8098 ± 0.0082	0.8198 ± 0.0117	0.8134 ± 0.0101	11.84M	Medium
Dual	0.8349 ± 0.0858	0.8412 ± 0.0846	0.8323 ± 0.0892	11.97M	Low
Tri-LF	0.7747 ± 0.0503	0.7786 ± 0.0531	0.7751 ± 0.0522	11.31M	Medium
Prompt-Qwen	0.4809 ± 0.0602	0.5021 ± 0.0460	0.3914 ± 0.0587	$\approx 18.46\text{M}$	High
MCSQ-Net	0.8521 ± 0.0929	0.8556 ± 0.0973	0.8547 ± 0.0955	$\approx 1.84\text{M}$	Moderately High

Note: the parameter count refers to the number of trainable parameters. Classical CNN baselines are estimated from the ResNet18 backbone and their corresponding fusion heads. Prompt-Qwen is estimated under the default LoRA setting ($r = 16$, $\alpha = 32$). MCSQ-Net is estimated under the three-modality setting used in the main experiments. Relative inference cost is assessed by jointly considering backbone size, number of input modalities, and whether autoregressive decoding is required.

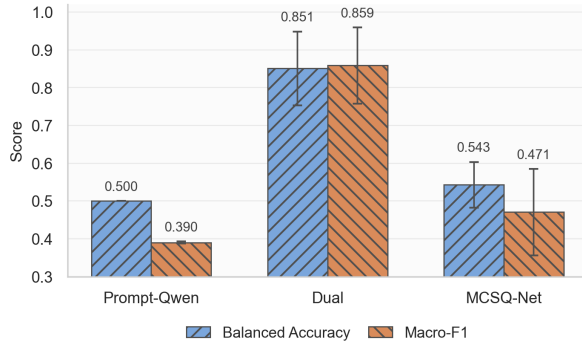


Fig. 5. Task-2 source-domain unseen-condition anomaly recognition results. Prompt-Qwen, the best classical baseline Dual, and MCSQ-Net are compared on Balanced Accuracy and Macro-F1. Error bars denote standard deviation over three folds.

constraint still improves the separation of abnormal patterns to some extent. Task-2 is therefore treated as a supplementary evaluation for distinguishing abnormal from normal samples.

Task-3 further tests robustness under external transfer. This task simultaneously introduces shifts in device imaging conditions, process-parameter ranges, and defect distribution, and therefore amplifies the model's reliance on condition-dependent superficial cues. Fig. 6 plots the performance trajectories of three representative methods from Task-1 to Task-3. Prompt-Qwen remains weak on both the source and external domains. The strongest classical transfer baseline, Tri-LF, drops from 0.7751 to 0.4236 in Macro-F1 and from 0.7786 to 0.4973 in Balanced Accuracy. In contrast, MCSQ-Net still achieves a Macro-F1 of 0.6067 and a Balanced Accuracy of 0.6359 on Task-3, with a substantially smaller cross-domain degradation. This result indicates that the causal consistency constraint helps reduce the interference of condition-related spurious cues on the fused representation.

This result also clarifies how the infrared modality contributes. Infrared information retains clear physical and statistical meaning in additive manufacturing monitoring [3], [4], but its benefit depends on the fusion mechanism and sample regime. In the classical transfer family, Tri-LF benefits from late fusion across three modalities, indicating that the way thermal information is exploited remains closely tied to the model family.

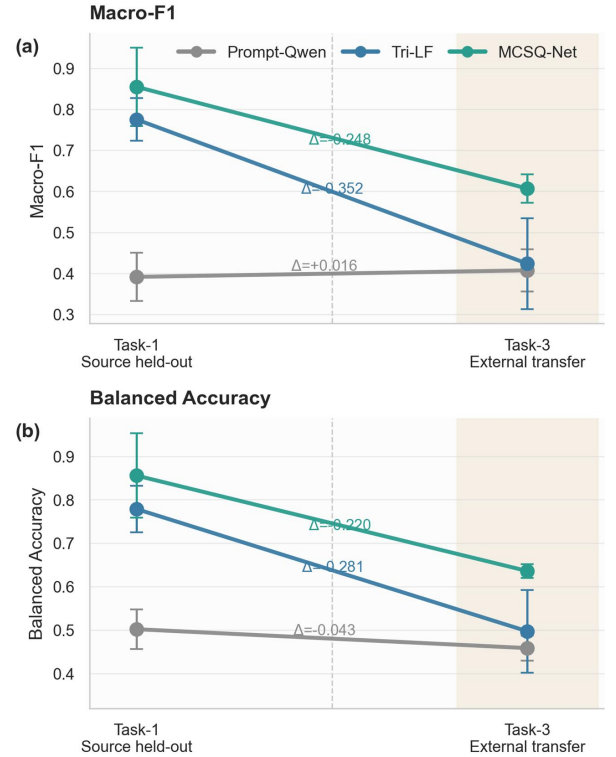


Fig. 6. Task-3 external-transfer defect category recognition results and cross-domain degradation. Top: Macro-F1. Bottom: Balanced Accuracy. Each trajectory connects the Task-1 source-domain result to the Task-3 external-transfer result. Error bars denote standard deviation over three folds, and the shaded region indicates the external target domain.

C. Ablation Studies

To analyze the source of MCSQ-Net's performance, this section conducts ablations from two perspectives: modality configuration and model structure. The results are summarized in Table V. The multimodal ablation shows that all modalities provide useful defect information, but their benefit depends on the specific fusion mechanism. The infrared modality alone still reaches a Macro-F1 of 0.6270, indicating independent discriminative value in thermal information. Among the ablated settings, the dual-camera input gives the most stable result, suggesting that complementary geometry and texture cues from two viewpoints are particularly suitable for the

TABLE V
ABLATION RESULTS OF MCSQ-NET

Ablation Type	Setting	Balanced Accuracy	Macro-F1
Input ablation	Camera 1	0.7714 $\pm$ 0.0771	0.7419 $\pm$ 0.1203
	IR	0.6508 $\pm$ 0.0708	0.6270 $\pm$ 0.0878
	Camera 1 & 2	0.8422 $\pm$ 0.1061	0.8402 $\pm$ 0.1048
	Camera 1 & 2 + IR	0.8169 $\pm$ 0.0659	0.8121 $\pm$ 0.0615
	Camera 1 & 2 + causal consistency	0.8556 $\pm$ 0.0973	0.8547 $\pm$ 0.0955
Model ablation	w/o causal consistency	0.8422 $\pm$ 0.1061	0.8402 $\pm$ 0.1048
	Full model	0.8556 $\pm$ 0.0973	0.8547 $\pm$ 0.0955
	Unfreeze the last two visual blocks	0.7979 $\pm$ 0.0332	0.7781 $\pm$ 0.0411

current setup. By contrast, adding infrared in a direct fusion manner does not further improve the result, implying that the current bottleneck lies not in whether infrared is introduced, but in how its thermal contribution is modeled. This interpretation is consistent with the transfer baseline results, where the strongest classical method adopts three-modality late fusion. The model ablation further verifies the role of the causal consistency constraint. Removing the constraint reduces Macro-F1 and Balanced Accuracy from 0.8547 and 0.8556 to 0.8402 and 0.8422, respectively. Unfreezing the last two visual blocks further lowers Macro-F1 to 0.7781, and also reduces Task-3 Macro-F1 from 0.6067 to 0.5808. These results indicate that the more effective strategy is not aggressive visual-backbone fine-tuning, but improving fusion-space stability while preserving the large-model prior.

VI. CONCLUSION

This paper studied multimodal defect recognition in metal additive manufacturing under unseen conditions, with a particular focus on how pretrained large-model priors can improve cross-condition stability. To this end, MCSQ-Net was proposed to learn stable defect representations through shared visual encoding, multimodal fusion, and a cross-condition causal consistency constraint. Experimental results show that the method outperforms Prompt-Qwen and classical multimodal baselines in both source-domain unseen-condition defect category recognition and external transfer evaluation. This indicates that prompting alone or conventional multimodal fusion is insufficient to cope with condition shift, whereas explicitly suppressing condition-related spurious cues in the fused representation is critical for generalization. Ablation studies further show that the causal consistency constraint consistently improves representation quality, while aggressively unfreezing the visual backbone weakens the original pretrained prior and leads to overfitting. Future work will investigate finer-grained fusion mechanisms for infrared thermal features and more conservative parameter-efficient adaptation strategies on the visual side, such as adapters or LoRA modules.

REFERENCES

- [1] B. Zhang, Y. Li, and Q. Bai, "Defect formation mechanisms in selective laser melting: A review," *Chinese Journal of Mechanical Engineering*, vol. 30, pp. 515–527, 2017.
- [2] X. Peng, L. Kong, H. An, and G. Dong, "A review of in situ defect detection and monitoring technologies in selective laser melting," *3D Printing and Additive Manufacturing*, vol. 10, no. 3, pp. 438–466, 2023.
- [3] D. Guillen, S. Wahlquist, and A. Ali, "Critical review of LPBF metal print defects detection: Roles of selective sensing technology," *Applied Sciences*, vol. 14, no. 15, p. 6718, 2024.
- [4] P. Wang *et al.*, "Towards intelligent defect detection in metal powder bed fusion: A review of in situ monitoring, data pre-processing, and machine learning," *Materials Science and Engineering: R: Reports*, vol. 167, p. 101112, 2026.
- [5] Z. Gu *et al.*, "AnomalyGPT: Detecting industrial anomalies using large vision-language models," arXiv:2308.15366, 2023.
- [6] Y. Li *et al.*, "Myriad: Large multimodal model by applying vision experts for industrial anomaly detection," arXiv:2310.19070, 2023.
- [7] S. Chen and Y. Liu, "Vision-language model-assisted active learning for high-accuracy defect detection in automotive bushing components," in *Proc. Int. Conf. Embodied Intelligence and Large Models*, 2025, pp. 9–14.
- [8] H. V. Nguyen, H. Park, N. Yoo, and J. Yang, "Resource-efficient fine-tuning of large vision-language models for multimodal perception in autonomous excavators," *Frontiers in Artificial Intelligence*, vol. 8, p. 1681277, 2025.
- [9] S. Zhao *et al.*, "OmniAD: Detect and understand industrial anomaly via multimodal reasoning," arXiv:2505.22039, 2025.
- [10] X. Jiang *et al.*, "AD-Copilot: A vision-language assistant for industrial anomaly detection via visual in-context comparison," arXiv:2603.13779, 2026.
- [11] Q. Wu *et al.*, "In-situ quality intelligent classification of additively manufactured parts using a multi-sensor fusion based melt pool monitoring system," *Additive Manufacturing Frontiers*, vol. 3, no. 3, p. 200153, 2024.
- [12] H. Liu *et al.*, "Inference of highly time-resolved melt pool visual characteristics and spatially-dependent lack-of-fusion defects in laser powder bed fusion using acoustic and thermal emission data," *Additive Manufacturing*, vol. 83, p. 104057, 2024.
- [13] T. Shen *et al.*, "Multi-source information fusion for enhanced in-process quality monitoring of laser powder bed fusion additive manufacturing," *Additive Manufacturing*, vol. 96, p. 104575, 2024.
- [14] J. Li *et al.*, "Transfer learning-based quality monitoring of laser powder bed fusion across materials," *Expert Systems with Applications*, vol. 252, p. 124150, 2024.